

DB_Julia: Exact Symbolic Verification of Detailed Balance for Lattice Monte Carlo

Abstract

DB_Julia decides, exactly and without sampling, whether a lattice Monte-Carlo update satisfies detailed balance with respect to a Boltzmann distribution, and does so *simultaneously for all values of the model's coupling parameters*. It intercepts every random draw the algorithm makes, reconstructs the exact symbolic transition probabilities by exhaustively enumerating the algorithm's decision tree, and verifies the detailed-balance identity as an exact polynomial identity over each region of coupling space. Every step uses exact rational arithmetic; anything that cannot be represented exactly is raised as a hard error rather than approximated, so the tool cannot return a false verdict.

1 Problem

A Markov chain on a finite state space with transition probabilities $T(s \rightarrow t)$ samples the target π if it satisfies *detailed balance*,

$$T(s \rightarrow t) \pi(s) = T(t \rightarrow s) \pi(t) \quad \text{for all states } s, t. \quad (1)$$

We take $\pi(s) \propto e^{-\beta E(s)}$ with a pairwise lattice energy

$$E(s) = \sum_{i < j} J_{t_i t_j}^{d^2(i,j)} \equiv \langle c(s), J \rangle, \quad (2)$$

i.e. E is *linear* in a finite vector of coupling parameters $J = (J_a)_{a \in \mathcal{A}}$ (one “atom” a per species pair and squared interaction distance; an optional field parameter is allowed). The integer vector $c(s)$ counts how many interacting pairs of each type occur in s . A state is a placement of typed particles on an $n \times n$ torus. The verifier confirms (1) for *every* pair (s, t) and *every* $J \in \mathbb{R}^{|\mathcal{A}|}$.

Input. The user supplies one MCMC step as a short Julia function `algorithm(rng, state)` together with `energy(state)`. All randomness goes through provided primitives (uniform choice, uniform integer, Metropolis acceptance, and a small algebra of symbolic acceptance thresholds for cluster moves). Geometry is queried only through translation-aware helpers.

2 Method

The pipeline has five exact stages.

1. State enumeration. All distinct placements of the species multiset on the torus are generated and cross-checked against the closed-form count $P(S, N) / \prod_k m_k!$, where $S = n^2$, N is the particle number and m_k the multiplicities. A mismatch is a hard error.

2. τ -augmented decision-tree enumeration. Each coordinate is carried as a τ -number $v + c_r \tau_r + c_c \tau_c$ (plus a flag marking any genuinely nonlinear τ term), where τ_r, τ_c are formal lattice-translation offsets and $v \in \mathbb{Q}$ is the true value (the algorithm runs at $\tau = 0$). Differences of positions cancel τ , so distances and occupancy tests are τ -free; squaring a coordinate produces a τ^2 term. A quantity is *translation invariant* iff its τ -coefficients vanish and it is untainted.

For a fixed seed state the algorithm is replayed against *every* finite bit string by a deterministic `BitSeqRNG`. A k -way uniform choice consumes $\lceil \log_2 k \rceil$ bits and multiplies an exact rational path probability (out-of-range patterns are rejected, reproducing exact rejection sampling); each acceptance test reads one bit and records a *symbolic* threshold rather than a number. Branching over both bit values enumerates the complete decision tree. Its leaves are the pairs $(t, w_{s \rightarrow t})$, and the weights sum to 1 by construction. Summing leaves with the same successor gives $T(s \rightarrow t)$ exactly.

Two structural facts are read off this tree. (i) If any recorded threshold, branch condition, or output coordinate carries a nonzero τ -coefficient, the update is *not* translation invariant and this is reported. (ii) Every output position is checked to be translation *covariant* (it shifts with the lattice); this licenses the orbit-reduction optimisation below and, when it fails, forces the safe all-states path.

3. Exact symbolic weights. A leaf weight is a rational coefficient times a product of acceptance factors. A Metropolis factor is the piecewise threshold $\min(1, e^{-\beta \langle \Delta, J \rangle})$; cluster moves (VMMC) need ratios of $(1 - e^{-\beta L})$ terms capped by a min. All such values are represented as exact *rational functions of exponential monomials*,

$$\text{Val} = \frac{\sum_v q_v e^{-\beta \langle v, J \rangle}}{\prod_k (1 - e^{-\beta \langle L_k, J \rangle})}, \quad q_v \in \mathbb{Q}, \quad (3)$$

with numerator a Laurent polynomial in the monomials and denominator a multiset of binomials. This class is closed under $+$, $-$, $\times$ and under division by a single binomial—exactly what the supported updates require. Anything outside it is a hard error.

4. Chambers of coupling space. Every branch condition is a linear inequality $\langle h, J \rangle \leq 0$; the finite set of such hyperplanes partitions $\mathbb{R}^{|A|}$ into open *chambers*, inside each of which every piecewise threshold takes a single branch and so resolves to a fixed Val. The chambers are enumerated by a breadth-first search over the hyperplane arrangement (adjacent chambers differ in one sign), where feasibility of each candidate sign pattern is decided by an *exact rational simplex*. Because the chamber-adjacency graph is connected, this search is complete given an exact feasibility oracle.

5. Detailed-balance residual. Using (2), $\pi(s) \propto e^{-\beta \langle c(s), J \rangle}$, so within a chamber the signed residual of (1),

$$R_{st}(J) = \sum_{\text{leaves } s \rightarrow t} w e^{-\beta \langle c(s), J \rangle} - \sum_{\text{leaves } t \rightarrow s} w e^{-\beta \langle c(t), J \rangle}, \quad (4)$$

is a rational function of exponential monomials. Multiplying through by the common $(1 - e^{-\beta L})$ denominator—nonzero on the open chamber—turns R_{st} into a single Laurent polynomial; detailed balance holds throughout the chamber iff *every* coefficient is zero. Coefficients are grouped by their exact rational exponent vector (handling half-integer exponents directly) and tested for vanishing in $\mathbb{Q}$. The algorithm passes iff $R_{st} \equiv 0$ for all communicating pairs and all chambers. Ergodicity is reported separately as reachability in the transition graph.

3 Soundness, determinism and performance

Exactness. All arithmetic is exact rational (`Rational{Int128}`). Julia’s rational arithmetic is overflow-checked, so an out-of-range computation throws and aborts rather than wrapping silently; the driver reports this as a hard error. The chamber test is an exact simplex with no tolerance, so no floating point enters the verdict.

Open-chamber completeness. Detailed balance is verified on the open chambers only. Since T and π are continuous in J , any violation on a positive-measure set already appears in an adjacent open chamber; measure-zero boundaries need not be tested separately.

Fail-loud. A path exceeding the depth bound, a non-integer or non-covariant coordinate, an empty or inverted random range, a min whose switch is not a genuine hyperplane, a division by a non-binomial, or a comparison that would leak an absolute position—each is a hard error, never a guess.

Determinism. There is no RNG anywhere; the bit-tree BFS, canonical orderings, and exact simplex are deterministic, and the optional multi-threaded mode is tested to reproduce the serial verdict, chamber count and transition graph bit-for-bit.

Performance. Three optimisations keep the cost down without affecting the result: translation-orbit reduction (BFS one representative per orbit and translate its leaves, valid because of the covariance check); per-pair projection onto each pair’s few active conditions; and hash-consing of thresholds so the model is built from unique weights. The verifier is exhaustive by construction and is intended for small representative systems; cost grows with the number of states and chambers.

Stage	Output	Exactness mechanism
State enumeration	all states	combinatorial cross-check
τ -BFS	leaves (t, w) , τ /covariance verdict	rational weights, formal τ
Weight algebra	$\text{Val} = \text{num} / \prod (1 - e^{-\beta L})$	closed rational-function class
Chambers	open regions of J -space	exact rational simplex
DB residual	per-pair, per-chamber	vanishing of $\mathbb{Q}$ -coefficients